

HTML - Elements & Attributes

1. HTML Elements Overview

- An HTML element is a component of an HTML document that defines the structure and content of a webpage.
- Elements are represented by **tags**, which are enclosed in angle brackets (`< >`).
- Most elements have an **opening tag** and a **closing tag**, with content in between.
 - Example: `<p>This is a paragraph.</p>`
- Some elements are **self-closing** and do not have a closing tag.
 - Example: ``

2. Basic HTML Elements

1. Headings:

- `<h1>` to `<h6>` : Define headings of different levels.
- Example: `<h1>Main Title</h1>`

2. Paragraphs:

- `<p>` : Defines a paragraph of text.
- Example: `<p>This is a paragraph.</p>`

3. Links:

- `<a>` : Creates a hyperlink.
- Example: `<a href="https://example.com">Visit Example</a>`

4. Images:

- `<img>` : Embeds an image.
- Example: ``

5. Lists:

- `<ul>` : Unordered list (bullet points).
- `<ol>` : Ordered list (numbered).
- `<li>` : List item.
- Example:

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

6. Divisions and Spans:

- `<div>` : Block-level container for grouping elements.
- `<span>` : Inline container for styling or grouping text.
- Example:

```
<div>This is a block container.</div>
<span>This is inline text.</span>
```

7. Forms:

- `<form>` : Creates a form for user input.

- `<input>` : Defines an input field.
- Example:

```
<form>
  <input type="text" placeholder="Enter your name">
</form>
```

8. Tables:

- `<table>` : Defines a table.
- `<tr>` : Table row.
- `<td>` : Table data (cell).
- Example:

```
<table>
  <tr>
    <td>Row 1, Cell 1</td>
    <td>Row 1, Cell 2</td>
  </tr>
</table>
```

9. Semantic Elements:

- `<header>` : Represents the header of a section or page.
- `<footer>` : Represents the footer of a section or page.
- `<section>` : Defines a section of content.
- `<article>` : Represents independent, self-contained content.
- Example:

```
<header>This is a header.</header>
<section>This is a section.</section>
<footer>This is a footer.</footer>
```

3. Self-Closing Elements

- Elements that do not have closing tags:
 - `<img>` : For images.
 - `<br>` : Line break.
 - `<hr>` : Horizontal rule.
 - `<input>` : Input fields.
 - Example: `<br>`

4. Attributes

- All HTML elements can have **attributes**
- The `href` attribute of `<a>` specifies the URL of the page the link goes to
- The `src` attribute of `<img>` specifies the path to the image to be displayed
- The `width` and `height` attributes of `<img>` provide size information for images
- The `alt` attribute of `<img>` provides an alternate text for an image
- The `style` attribute is used to add styles to an element, such as color, font, size, and more
- The `lang` attribute of the `<html>` tag declares the language of the Web page

- The `title` attribute defines some extra information about an element

1. `style` Attribute

- The `style` attribute is used to apply **inline CSS styles** directly to an HTML element.
- It allows you to control properties like color, font, size, alignment, and more.
- Example:

```
<p style="color: red; font-size: 20px;">This is a styled paragraph.</p>
```

- **Common Properties:**

- `color` : Sets text color.
- `font-size` : Controls the size of the text.
- `background-color` : Sets the background color of an element.
- `text-align` : Aligns text (left, right, center, justify).
- `margin` and `padding` : Control spacing around elements.

2. `lang` Attribute

- The `lang` attribute specifies the **language of the content** within an element.
- It is commonly used in the `<html>` tag to declare the language of the entire webpage.
- Helps screen readers and search engines understand the language of the content.
- Example:

```
<html lang="en">  
  <body>  
    <p>This page is in English.</p>  
  </body>  
</html>
```

- **Language Codes:**

- `en` for English.
- `es` for Spanish.
- `fr` for French.
- `de` for German.

3. `title` Attribute

- The `title` attribute provides **additional information** about an element.
- When a user hovers over the element, the text in the `title` attribute is displayed as a tooltip.
- Useful for adding context or descriptions to links, images, or other elements.
- Example:

```
<a href="https://example.com" title="Visit Example Website">Click here</a>  

```

4. `href` Attribute

- The `href` attribute is used in the `<a>` tag to specify the **destination URL** of a hyperlink.
- It can link to external websites, internal pages, or specific sections within a page.

- Example:

```
<a href="https://www.w3schools.com">Visit W3Schools</a>
<a href="#section1">Go to Section 1</a>
```

5. src Attribute

- The `src` attribute is used in elements like `<img>`, `<script>`, and `<iframe>` to specify the **source file**.
- For images, it defines the path to the image file.
- Example:

```

<script src="script.js"></script>
```

6. alt Attribute

- The `alt` attribute provides **alternative text** for images.
- It is displayed if the image fails to load and is used by screen readers for accessibility.
- Example:

```

```

7. width and height Attributes

- The `width` and `height` attributes specify the **dimensions** of an image or other elements.
- Values can be in pixels (`px`) or percentages (`%`).
- Example:

```

```

8. Other Common Attributes

- `id` : Assigns a unique identifier to an element.

```
<div id="header">This is the header.</div>
```

- `class` : Assigns one or more class names to an element for styling or scripting.

```
<p class="highlight">This is a highlighted paragraph.</p>
```

- `target` : Specifies where to open a linked document (e.g., `_blank` opens in a new tab).

```
<a href="https://example.com" target="_blank">Open in New Tab</a>
```